

Homework 2

Total Marks: 40

Deadline: Mar 29, 2026, at 11:55 pm

General Instructions: You may partner with up to 3 other students (a group of up to 4) to submit a single write-up in a single group. We encourage discussion with other students, even if they are not in your group. Each student must develop a solid understanding of all the questions, even when discussing and collaborating with others. Merely dividing the problems among group members and putting those together before submission will severely impact your learning, and we **strongly advise against it**. Furthermore, while it is okay to use online resources, including **Generative AI** such as **ChatGPT**, for learning purposes, **blatant copying** or **mindlessly querying** the Generative AI is **forbidden**. If there are any confusions or questions, post those on the WhatsApp group or see your TA or Instructor.

Submission Instructions: Typed or hand-written solutions (only in PDF format) can be submitted on LMS before the deadline. Email submissions WILL NOT be accepted. *Only one submission per group is needed; the first page must clearly mark the names and student IDs of all group members.*

Question 1 [5 marks] In Bitcoin, each user's balance is represented using the hash of a public key. A user's name or identity is not stored on the blockchain. An adversary may still use the transaction history associated with a public key in order to arrive at the identity of a user. Give an example of how this is possible and devise two strategies that may be used to protect the anonymity of a user.

Question 2 [6 marks]

- a) Explain why strictly more than half of the network resource guarantees eventual chain overtake.
- b) Assume the network consensus currently considers Block#271 to be the latest block. A malicious miner controlling 51% of the network has the ability to create a private fork. Describe a double-spend attack that this miner could use against an exchange that credits deposits only after k confirmations (the exchange deposits cash into the user's account in exchange for cryptocurrency after k blocks have been mined after it). Be explicit about the miner's actions and the point at which the exchange would be tricked.
- c) Give reasons why 51% attacks are still rare in practice.

Question 3 [5 marks] Imagine that a blockchain is about to activate a soft fork that reduces the block size limit. Most miners have upgraded, but Aqib controls 51% of the network's compute power and refuses to upgrade.

- a) Describe what will happen to the chain if Aqib mines blocks that violate the new, smaller block size limit. Will the chain split temporarily, permanently, or not split at all?
- b) Describe what will happen in the other scenario where Aqib, with 51% of compute power upgrades, while all the other nodes stay non-updated. Will the chain split temporarily, permanently, or not split at all? Will he be able to double-spend there?

Question 4 [5 marks] When a soft fork is invoked, does it need the majority of nodes to upgrade to ensure that a single chain emerges? Why or why not?

Question 5 [5 marks] Read up on "Grover's algorithm". Will the existence of this algorithm (and similar algorithms) affect cryptographic hash functions such as SHA256? What about symmetric key encryption, such as AES-128?

Question 6 [6 marks]

- a) In simple words, explain what a soft fork and a hard fork are, focusing on what happens to nodes that do not upgrade.
- b) Give one example of a rule change for each type of fork (other than block size or mining reward) and explain why it would be a soft or a hard fork.

Question 7 [4 marks] Suppose Aqib is a malicious miner who controls 51% of the network's compute power. This means he has complete control of the blocks he mines. As we have seen previously, he maliciously uses this power to double-spend his cryptocurrency. Is there any other way he can exploit his control over the mined blocks? Can he change the outputs of the fair transactions to his own address?

Question 8 [4 marks] Explain the concept of non-repudiation in digital signatures. Why is this property critical for cryptocurrencies like Bitcoin? How does Bitcoin ensure it?